



1.11

Equivalent Representations of Polynomial and Rational Expressions

Core-to-Challenge Worksheet and AP Practice

AP Precalculus - Unit 1 | ECHS Mathematics | Teacher: Mohammad Abu-Ghuwaleh

Student Information

Name: _____ Class: _____ Date: _____

Directions

Show mathematical evidence. Retain domain restrictions, label graphs, include units, and write complete interpretations. Calculator use is permitted only when the item explicitly requires approximation, regression, or numerical solving.

Readiness

1. Write one precise sentence explaining this principle: Standard form highlights degree and leading coefficient.

2. Write one precise sentence explaining this principle: Factored form highlights zeros, multiplicities, cancellations, and sign changes.

3. Write one precise sentence explaining this principle: Division form highlights polynomial asymptotes and remainders.

4. Write one precise sentence explaining this principle: Equivalence is algebraic; equality as functions also requires matching domains.

Core

5. Rewrite $x^3 - 4x^2 - x + 4$ in factored form.

6. Divide $2x^3 + 3x^2 - 5$ by $x - 1$.

7. Why is $\frac{x^2-1}{x-1}$ not the same function as $x + 1$?

8. State what the formula $P(x) = a_nx^n + \cdots + a_0$ tells you and name any required restriction.

9. State what the formula $P(x) = a \prod (x - r_i)^{m_i}$ tells you and name any required restriction.

10. State what the formula $\frac{P}{Q} = S + \frac{A}{Q}$ tells you and name any required restriction.

Medium

11. Rewrite $x^3 - 9x$ into a form that reveals zeros or restrictions.

12. Rewrite $x^4 - 16$ into a form that reveals zeros or restrictions.

13. Rewrite $2x^3 + x^2 - 8x - 4$ into a form that reveals zeros or restrictions.

14. Rewrite $\frac{x^2+5x+6}{x+2}$ into a form that reveals zeros or restrictions.

15. Rewrite $\frac{x^3+1}{x+1}$ into a form that reveals zeros or restrictions.

Hard

16. Divide $x^4 - 3x^2 + 2x - 5$ by $x - 2$.

17. Use the binomial theorem to find the coefficient of x^3 in $(2x - 1)^6$.

18. Which form of $P(x) = x^3 - 4x$ best reveals end behavior, zeros, and y-intercept?

19. Explain why synthetic division by $x-c$ evaluates $P(c)$.

Difficult

20. Find k so $x^3 + kx^2 - 4x - 4k$ has factor $x-2$ and factor $x+2$.

21. Rewrite $\frac{x^3 - 2x^2 + 5}{x^2 - 1}$ in division form and use it to identify asymptote.

22. Find all x for which expanded forms $(x - 1)^4$ and $x^4 - 4x^3 + 6x^2 - 4x + 1$ define different functions.

Challenge / HOT

23. Prove that two polynomials equal at infinitely many distinct real inputs are identical.

24. Find a polynomial P such that $P(x + 1) - P(x) = 4x^3 + 6x^2 + 4x + 1$.

AP-Style Multiple Choice

25. Factored form most directly reveals
- (A) leading coefficient only
 - (B) zeros and multiplicities
 - (C) all residuals
 - (D) log base

26. Division form most directly reveals
(A) polynomial asymptote
(B) domain of a polynomial
(C) complex conjugates only
(D) finite differences
27. Remainder on division by $x-3$ equals
(A) $P(0)$
(B) $P(1)$
(C) $P(3)$
(D) $P(-3)$
28. $(x+2)^3$ expanded has x^2 coefficient
(A) 2
(B) 6
(C) 8
(D) 12
29. Cancelling a factor requires retaining
(A) leading sign
(B) original domain restriction
(C) new zero
(D) new coefficient
30. $x^2 - 5x + 6$ factors as
(A) $(x-1)(x-6)$
(B) $(x-2)(x-3)$
(C) $(x+2)(x+3)$
(D) prime

AP-Style Free Response

31. For $R(x) = \frac{x^3 - 4x^2 - x + 4}{x^2 - 1}$: (a) factor completely; (b) simplify with restrictions; (c) identify holes/asymptotes/zeros; (d) write division form of the simplified expression.

32. Use the binomial theorem to expand $(x-2)^5$, then use the expansion to find the fourth finite difference for unit steps and explain why it is not constant.

Reflection

Identify one item where a domain restriction, unit, assumption, or representation changed your reasoning. Explain in 2-3 sentences.

Original ECHS questions. Selected concepts are informed by Pearson and Holt Precalculus; no textbook exercises are reproduced.